DTU

34346 Networking Technologies and Application Development for IoT

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Beam Buddy : IoT-Enabled Smart Bike Light

Overview

Problem statement

Bike lights are a critical safety component for any cyclist, but they are often unreliable (run out of battery), easily stolen, or easy to forget.

Our Solution

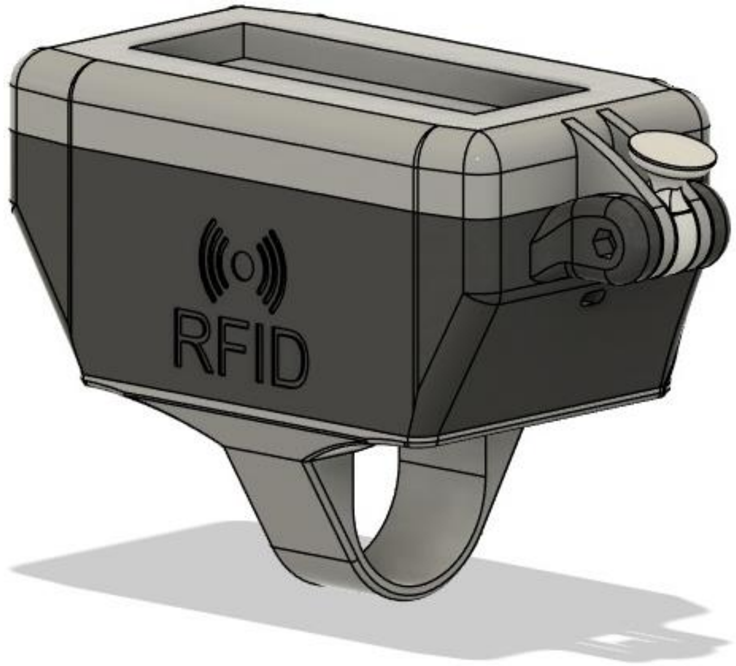
- front and rear light
- turns on automatically with motion/darkness
- theft alarm with RFID card for locking/unlocking the bike when parked
- solar charging + audible warning for low battery
- Web interface for visualizing operation mode, last seen location, and battery status

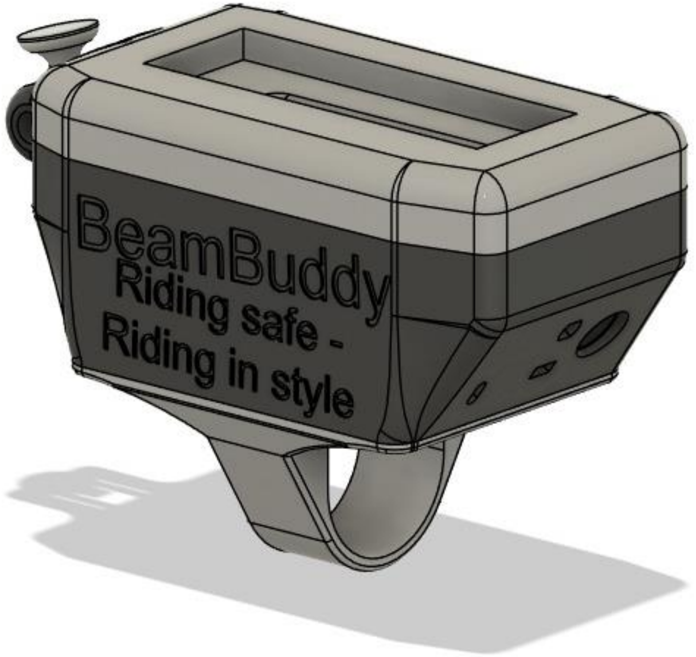
Design

Design- prototype

Based on an HT-CT62 with features:

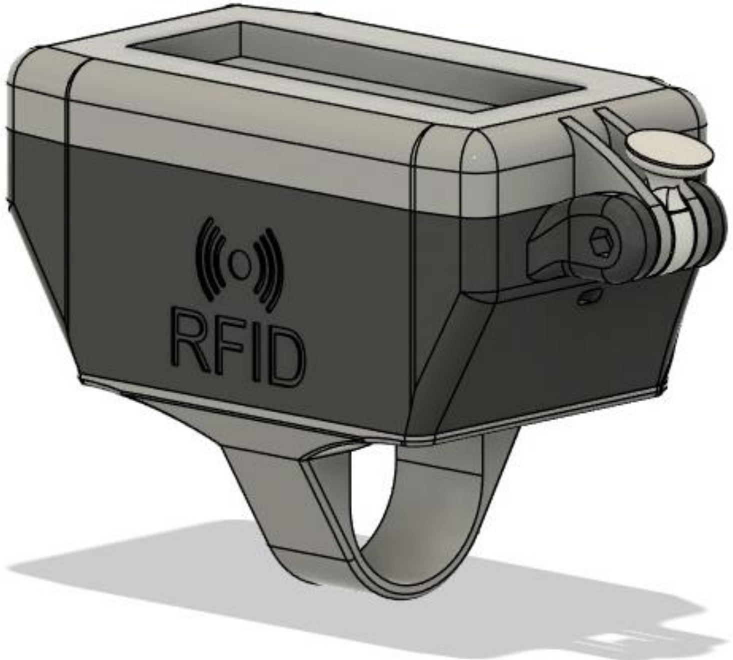
- Automatic lights w. overwrite button
- GPS-tracking using WiFi and GNSS
- Movement and light sensors
- RFID lock
- Theft alarm system
- LoRaWAN wireless communication
- App communication
- Solar & 5V charging capabilities
- UV secure casing

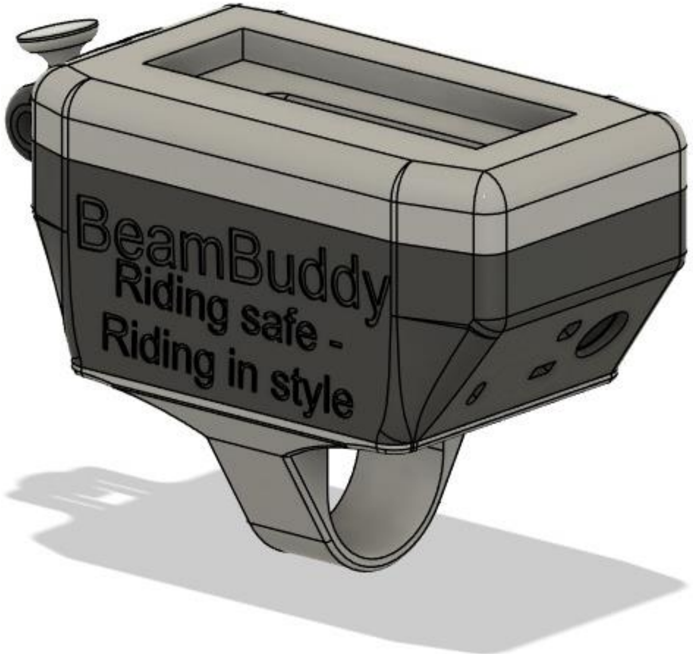




Enclosure

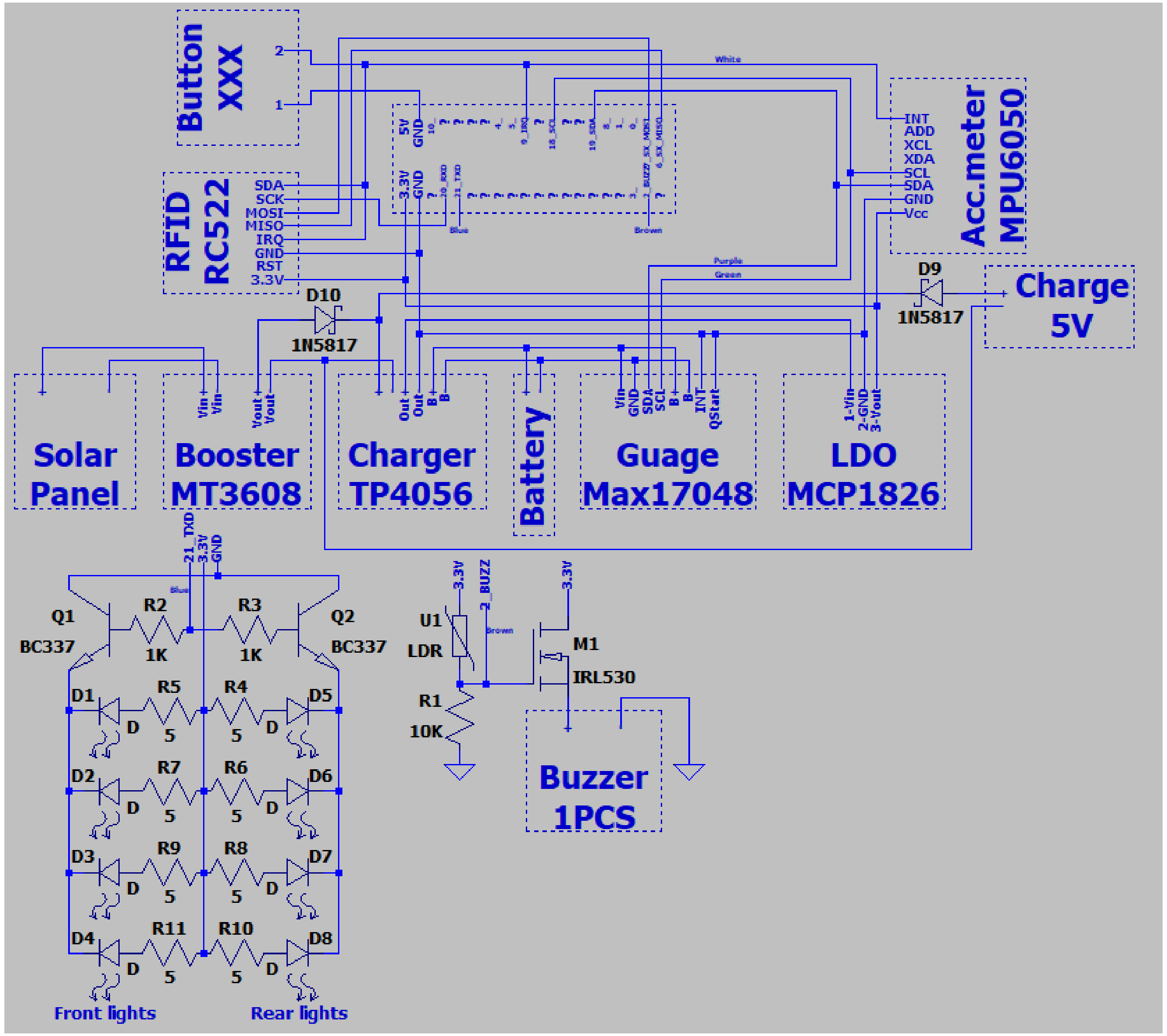
- Waterproof capable
- UV resistant
- Secure anti-theft mounting
- 3D Printed
- PETG casing





Hardware Design

- Heltec HT-CT62 Development Board (ESP32 framework with an SX1262)
- MT3608 Booster
- TP4056 Charging circuit
- ICR18650-26J battery (3.7V 2600mAH)
- MAX17048 Battery Guage
- MCP1826 LDO
- RC522 RFID read/writer
- MPU6050 Accelerometer
- 1N5817 Diode protection
- BC337 BJT controlling lights
- IRL530 MOSFET controlling buzzer

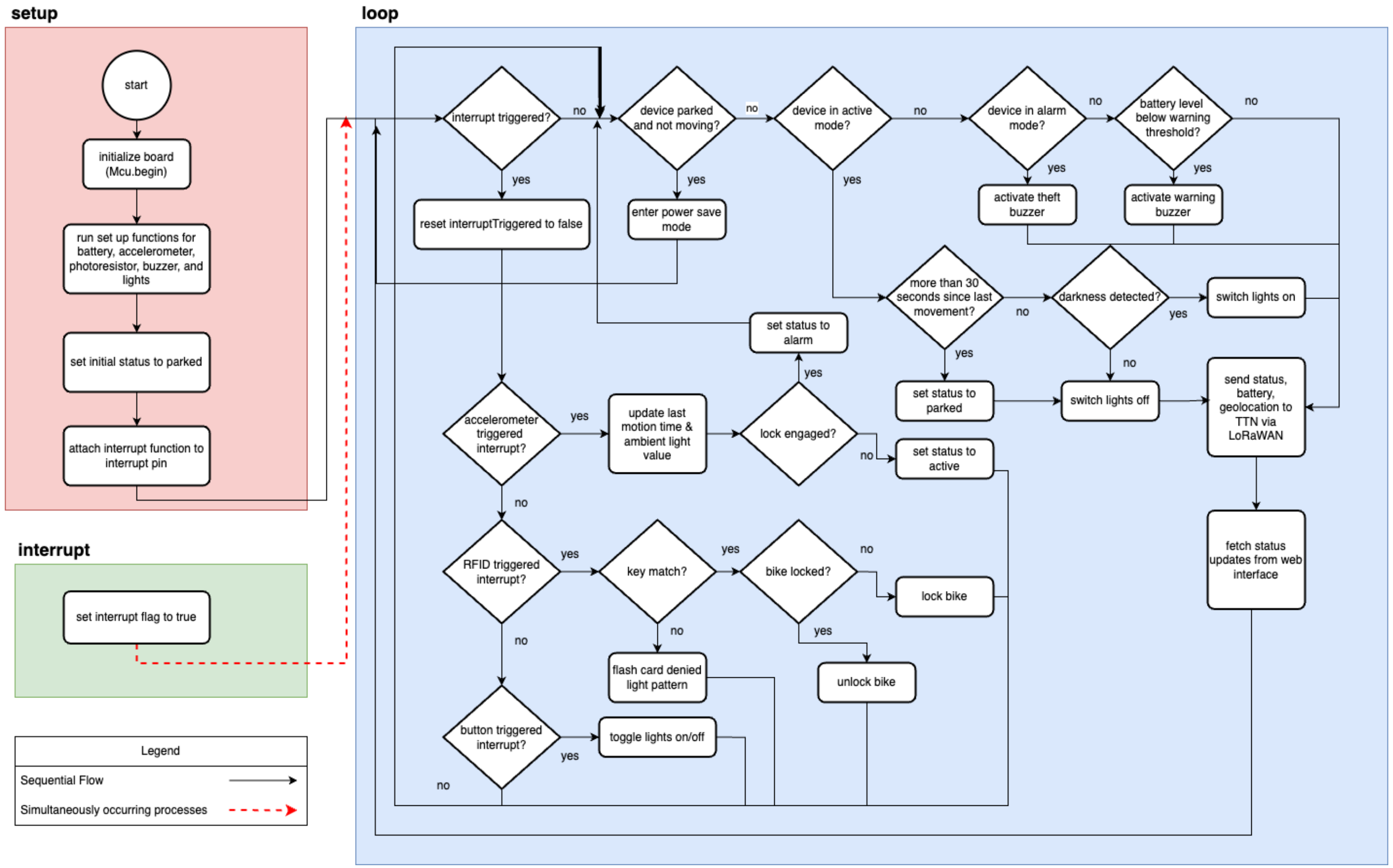


Software Design

Operating Modes:

- **Storage** (deep sleep)
- **Parked** (locked or unlocked)
- **Alarm**
- **Active**

Single interrupt triggered by accelerometer, RFID, or button



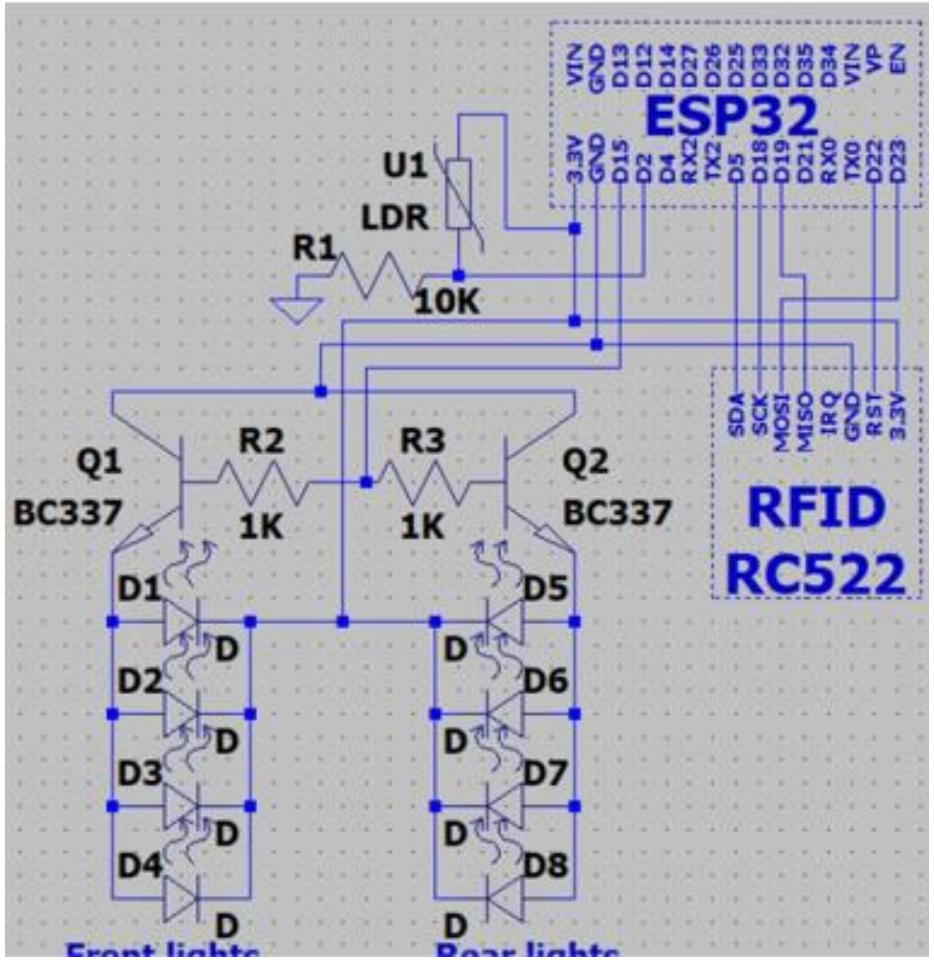
Implementation

LED Light

- Automatic turn on/off with motion/light
- Manual overwrite function
- Power efficient visibility light
- Front and rear light
- Angle adjustable front light

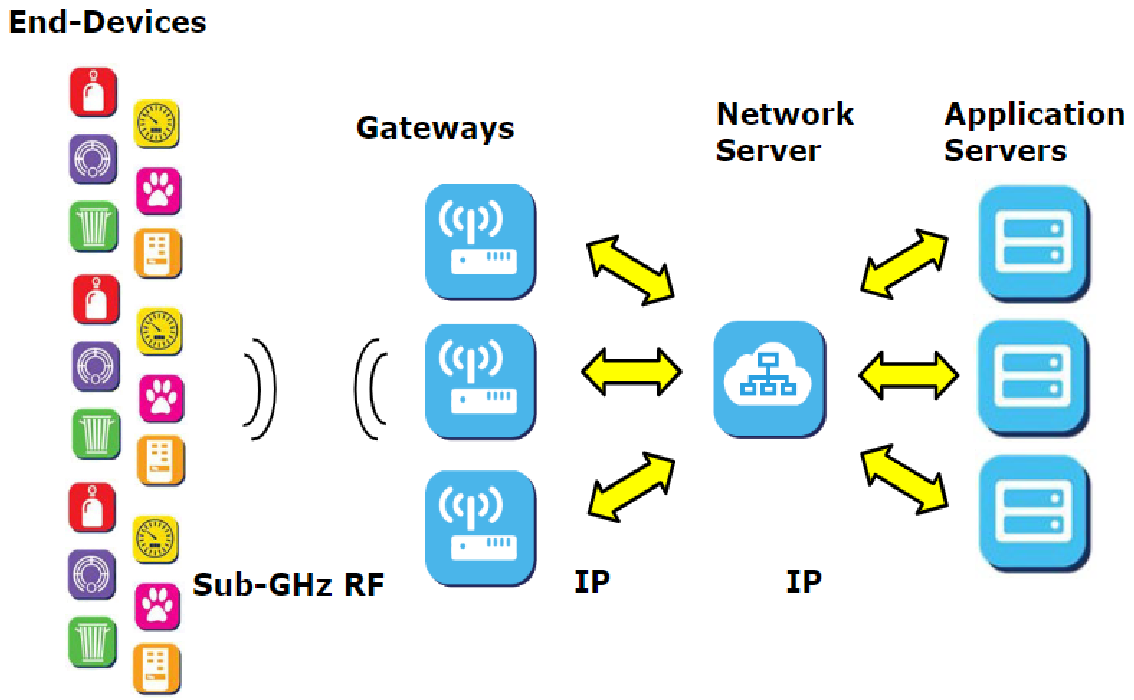
RFID Lock + Theft Alarm

- Rolling FOB – capable
- Multiple Tags
- Locking, unlocking & error light indication
- Movement activated theft alarm
- Audible alarm



Network technology - LoRaWAN

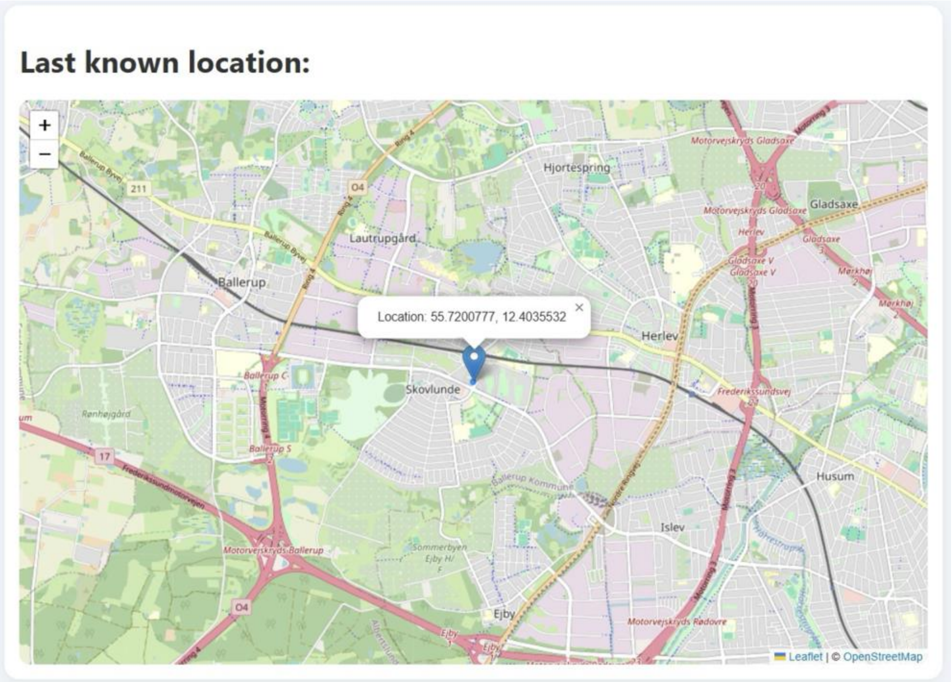
- **Advantages:**
  - Long range
  - Low power consumption
  - Cost effective + scalable
  - Public network (TTN)
- **Disadvantages:**
  - Inconsistent coverage
  - susceptible to interference



Geolocation – WiFi Scanning

Method:

- The ESP32 board scans for nearby WiFi networks.
- MAC addresses are filtered and compressed.
- Payload of valid MAC addresses + RSSI sent to TTN via LoRaWAN
- Webhook forwards payload to application server (deployed on Railway).
- Google API queries MAC address data and returns geolocation.



Network technology – Web Integration

Comparison of web integration opportunities on TTN.

Webhook fits the requirements best.

Feature	Webhook	WebSocket
Connection	One-time HTTP POST	Persistent TCP connection
Direction	One-way (push only)	Two-way (send + receive)
Duplex	❌ No (half-duplex or push)	✅ Yes (full-duplex)
Use Case	Event notification	Real-time communication

Network technology – Application server

Backend:

Python

- Parses the JSON structure.
- Decompresses the data.
- Saves the latest data locally in a JSON file to track recent activity.
- Utilizes Google Geolocation API to get position

Frontend:

JavaScript and HTML and CSS.

Power supply

- Battery function
  - Returns a Byte
  - 7LSB battery percent
  - 1MSB low voltage alert

Module	Description	Notable features
ICR18650-26J	Lithium-ion cell	• 3.63V Nominal voltage • 4.2V charge voltage • 2.75V cut-off voltage
TP4056	Charging circuit	• 4.2V charge voltage • 6.5V max input voltage • Micro-USB charging
MCP1826S	3v3 LDO regulator	• 0.25V typical dropout voltage • 0.4V max dropout voltage
	Solar cells	• 6V max output voltage
MT3608	Step-up converter	• 2V min input voltage • 4.2V step-up voltage
MAX17048	Fuel gauge	• ±7.5mV Voltage measurement accuracy • Programmable alerts • I2C interface

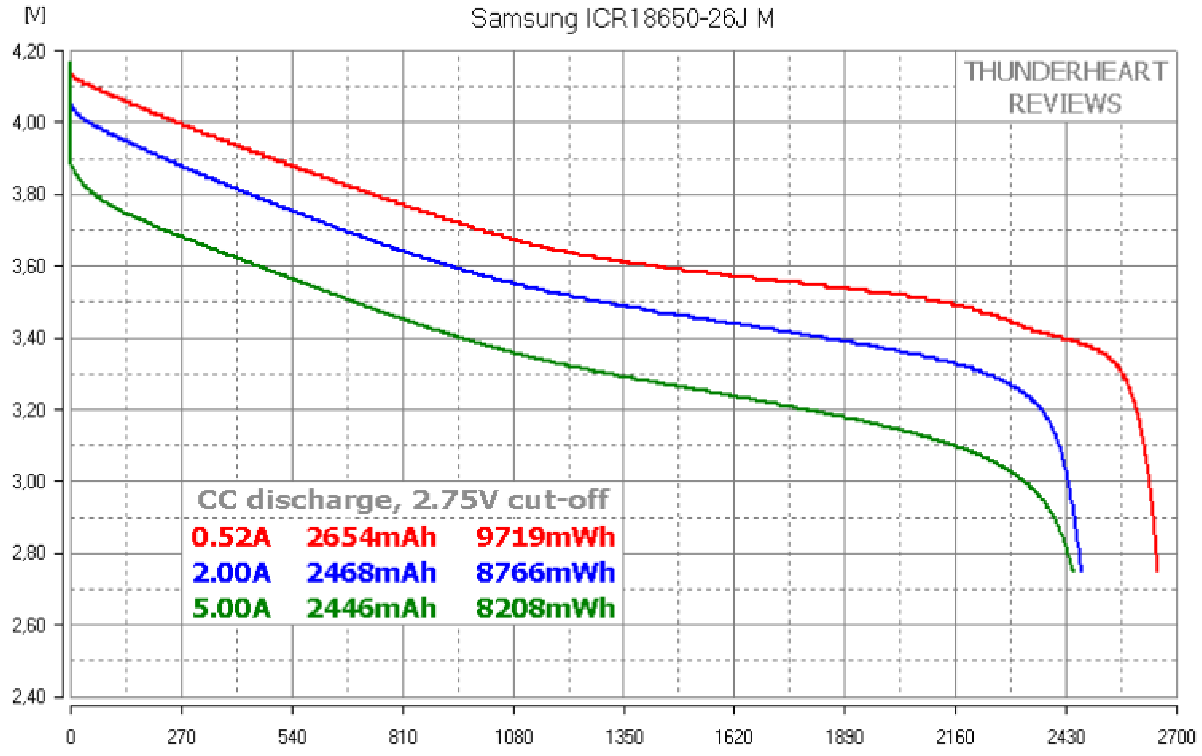


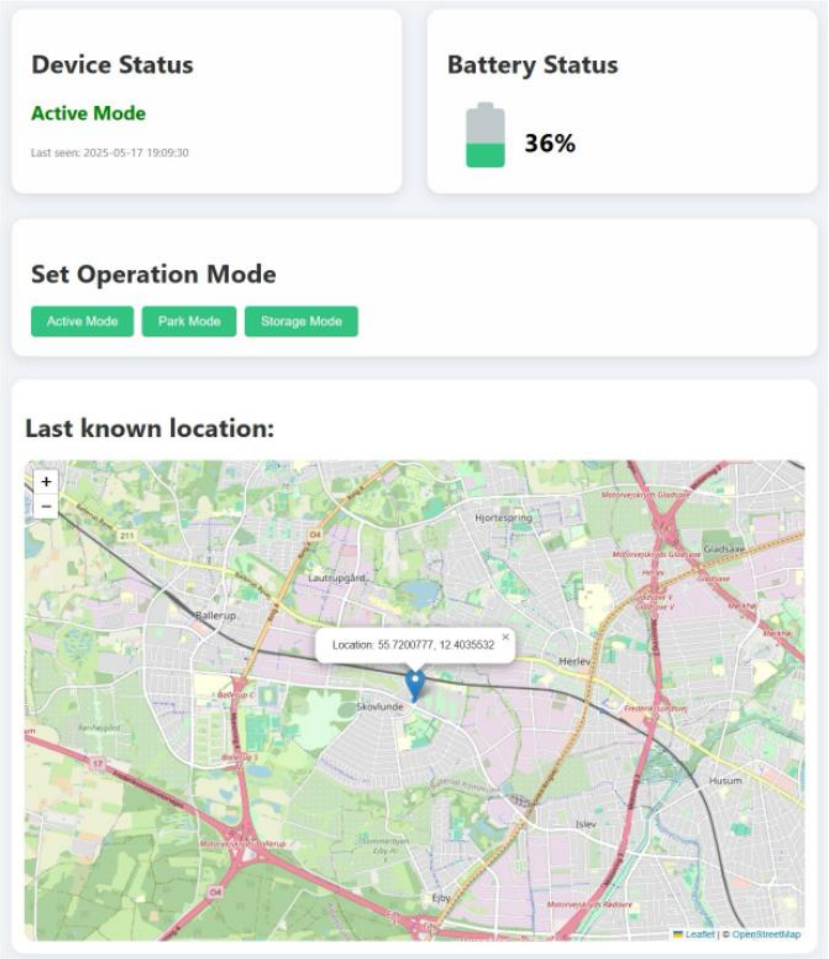
Figure 1: Samsung 26J capacity test with V on y-axis and mAh on x-axis[1].

Results

Final Product

Graphical User Interface

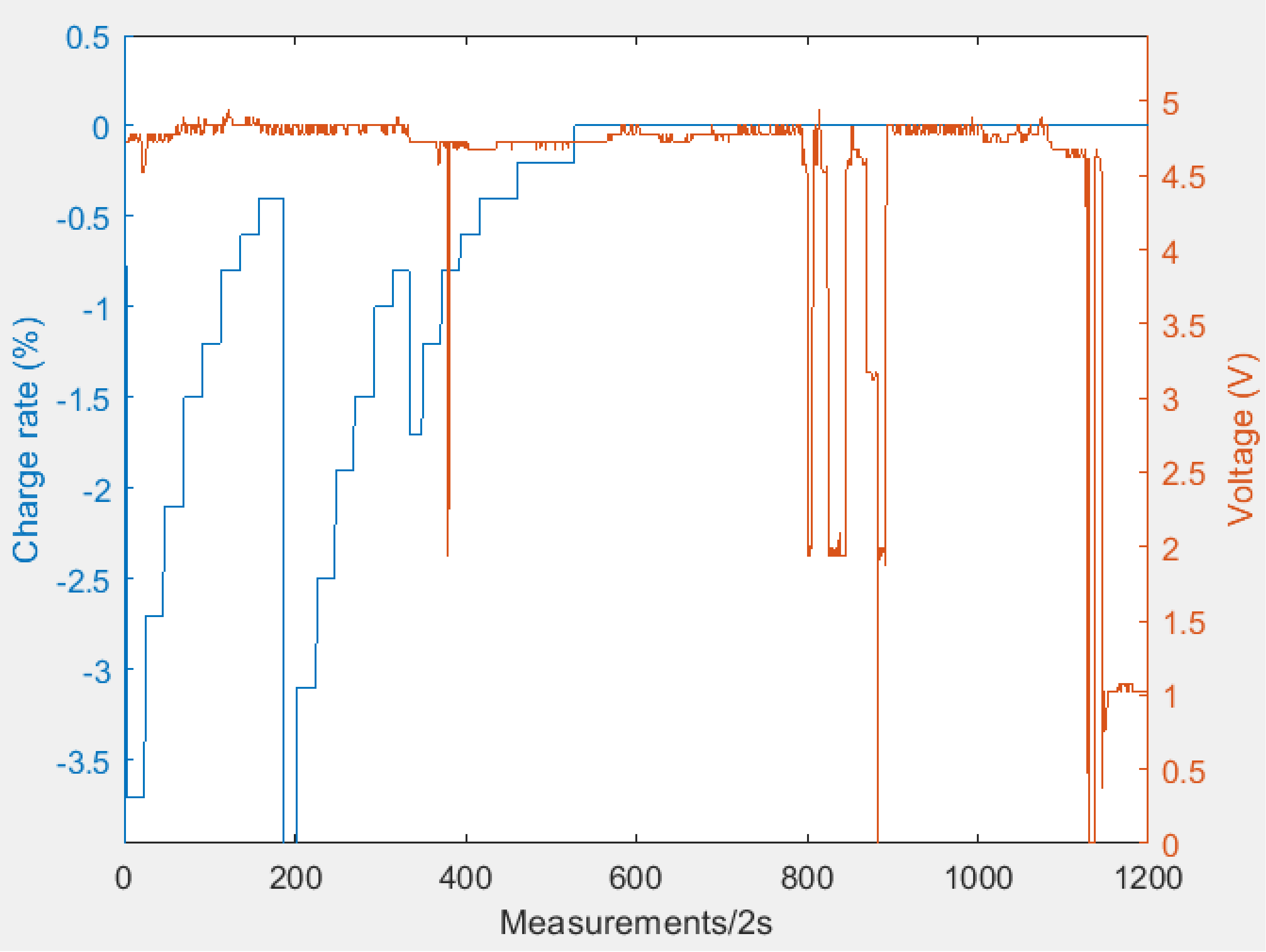
- Displays battery status, last known location, and current operation mode
- Stores last known data in a JSON file
- Option to set operation mode
- Works on mobile and desktop



Conclusions

Future Improvement: Solar charging

- Problems:
  - Produces 5V in direct sunlight.
  - At charging circuit voltage too close to charging voltage.
  - Step up converter set too low 4.25V, and not suitable for fluctuating inputs.
- Solutions:
  - More efficient solar panels with a buck converter.
  - Adjusting the step-up converter or removing it.
  - Dedicated solar charging circuit.



Future Improvement: GNSS for Geolocation

Dual mode: WIFI scanning and GNSS module.

- In urban area send MAC address and RSSI to Google API to get geolocation.
- In rural area when no WIFI available, use GNSS.(Not in the final code due to restrictions)

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Summary

- Complete Successes
  - 3D design (compact enclosure) w/front and rear light
  - Low power geolocation w/wifi scanning
  - Lights responsive to motion + ambient light
  - Web interface displaying current battery, status and last known location

- Partial Successes
  - RFID anti theft lock (needs further testing)
  - Set status from web interface w/LoRaWAN downlink (unreliable)

- Limitations
  - Brownout issue causing reset
  - Sparse LoRaWAN network coverage
  - No deep sleep mode